

# Erdős Problem #597

EP-597

L1 - TRACTABLE

PARTIALLY SOLVED

Graph Theory

## ORIGINAL PROBLEM STATEMENT

Let  $G$  be a graph on at most  $\aleph_1$  vertices which contains no  $K_4$  and no  $K_{\aleph_0, \aleph_0}$  (the complete bipartite graph with  $\aleph_0$  vertices in each class). Is it true that  $\omega_1^2 \rightarrow (\omega_1 \omega, G)^2$ ? What about finite  $G$ ?

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Archive: <https://www.unsignedmath.com/problems?difficulty=1>  
Dataset: <https://huggingface.co/datasets/ulamai/UnsolvedMath>